# North Sea Oil Platform Drill Tower Construction

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## 1. Executive Steering Committee Status Briefing

**Project Name**: Offshore EPC Platform Drill Tower Project

**Status Date**: August 31, 2026 (Month 8 / Status Week 36)

**Report Type**: Monthly Executive Steering Committee Status Briefing

**Author**: Frank Ellingsen, Lead Project Controller

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### 🚦 OVERALL PROJECT HEALTH DASHBOARD: RED (CRITICAL)

| Overall RAG Status | Cost Performance ($CPI$) | Schedule Performance ($SPI_t$) | Critical Ratio ($CR$) | Budget Burn Out Month | Predicted Outturn Cost ($EAC_1$) | Predicted Commercial COD |
|---|---|---|---|---|---|---|
| 🚨 **CRITICAL OVERRUN** | **0.7483** (Red) | **0.9250** (Amber) | **0.6402** (Red) | **Month 9 (Sep 2026)** | **$35,413,604** (+$8.91M) | **Jan 31, 2027** (+31 Days) |

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### 1. KEY PERFORMANCE INDICATORS & EVM METRICS SUMMARY

- **Baseline Budget ($BAC$)**: **$26,500,000**
- **Earned Physical Completion ($EV / BAC$)**: **66.19%** ($EV = \$17,540,000$)
- **Planned Baseline Completion ($PV / BAC$)**: **77.36%** ($PV = \$20,500,000$)
- **Actual Costs Incurred ($AC$)**: **$23,440,000**
- **Cost Variance ($CV = EV - AC$)**: **-$5,900,000** ($-33.6\%$ cost overrun to date)
- **Schedule Variance ($SV = EV - PV$)**: **-$2,960,000** ($-14.4\%$ work lag to date)
- **Earned Schedule ($ES$)**: **7.40 Months** ($SV_t = -18.2\text{ Days}$, $SPI_t = 0.9250$)

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### 2. OUTTURN FORECASTS & MONTE CARLO P90 RISK ANALYSIS

| Outturn Scenario | Forecast Cost ($EAC$) | Cost Overrun ($VAC$) | Completion Date | Schedule Delay | Required Risk Contingency |
|---|---|---|---|---|---|
| **Deterministic Most Likely ($EAC_1$)** | **$35,413,604** | **-$8,913,604** | **Jan 31, 2027** | **+31 Days** | Base Forecast ($0.00) |
| **Monte Carlo P50 (Median)** | **$34,060,783** | **-$7,560,783** | **Feb 27, 2027** | **+57 Days** | Base Forecast ($0.00) |
| **Monte Carlo P90 (90% High Confidence)** | **$35,815,202** | **-$9,315,202** | **Mar 14, 2027** | **+72 Days** | **+$401,598 Reserve** |

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### 3. PRIMARY COST & SCHEDULE VARIANCE DRIVERS

1. **Egersund Mast Assembly Yard Rework ($WBS\text{ }1.3.2$)**: Structural pipe out-of-tolerance requiring 24/7 NDT welding. Accounts for **$2.40M direct overrun** ($26.9\%$ of total variance).

2. **Time Delay Overhead Spread ($SPI_t = 0.9250$)**: Extending project execution past Dec 31, 2026 adds **$3.81M in time-based site overhead** ($42.8\%$ of total variance).

3. **Heavy Lift Vessel Weather Standby ($WBS\text{ }1.4.1$)**: Marine vessel mobilization delayed into autumn sea states (*Heerema Sleipnir*), incurring **$1.30M standby cost** ($14.6\%$ of variance).

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### 4. CASH RUNWAY & LIQUIDITY ALERT

- **Remaining Baseline Capital ($BAC - AC$)**: **$3,060,000**
- **Average Monthly Cash Burn Rate**: **$2,930,000 / month**
- **Budget Burn Out Date**: **Month 9 (September 2026)** — 100% of the $26.50M baseline budget will be depleted.
- **Urgent Action**: Secure executive approval for a **+$8.91M overrun credit line before September 15, 2026**.

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### 5. RECOMMENDED STEERING COMMITTEE DECISIONS

1. **Approve Additional Funding Line**: Authorize **$8.91M** overrun credit line to cover Months 9–13 operations.

2. **Implement Fixed-Fee Yard Agreement**: Cap Egersund yard billing under a fixed-fee labor agreement for remaining assembly.

3. **Execute Critical Path Recovery**: Shift 15 calendar days from hook-up testing to preserve **January 31, 2027 COD**.

## 2. Earned Value & Outturn Predictions

**Project Name**: Offshore EPC Platform Drill Tower Project

**Status Date**: August 31, 2026 (Month 8 / Status Week 36)

**Author**: Frank Ellingsen, Lead Project Controller

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### Executive Summary & Outturn Prediction Matrix

At Status Month 8, the project has achieved **66.19% physical completion** ($EV = \$17.54\text{M}$) against a cumulative budget spent of **$23.44M** ($AC$) and a baseline plan of **$20.50M** ($PV$).

Based on standard Earned Value Management (PMI EVM Practice Standard) and Earned Schedule ($ES$) statistical modeling, the project **will not achieve its original $26.50M budget or December 31, 2026 completion date**.

#### 📊 Summary of Final Outcome Scenarios

| Forecast Scenario | Metric | Outturn Cost ($EAC$) | Variance ($VAC$) | Completion Date | Schedule Delay ($TV$) | Probability | Primary Driving Assumptions |
|---|---|---|---|---|---|---|---|
| **Best Case (Idealistic)** | $EAC_2$ | **$32.40M** | -$5.90M (-22.3%) | **Jan 15, 2027** | +15 Days | 5% | Future work performed strictly at 100% budget efficiency ($CPI_{future} = 1.00$). |
| **Most Likely (Standard EVM)** | $EAC_1$ | **$35.41M** | **-$8.91M (-33.6%)** | **Jan 31, 2027** | **+31 Days** | **70%** | **Remaining work performed at current cumulative efficiency ($CPI = 0.75$, $SPI_t = 0.925$).** |
| **Worst Case (Composite)** | $EAC_3$ | **$37.43M** | -$10.93M (-41.3%) | **Feb 15, 2027** | +46 Days | 25% | Cumulative cost & schedule compounding ($CPI \times SPI = 0.64$). |

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### 1. Cost Predictions ($EAC$ & $VAC$)

#### Status Metrics at Month 8 ($M8$)

- **Budget at Completion ($BAC$)**: $\$26,500,000$
- **Planned Value ($PV$)**: $\$20,500,000$
- **Earned Value ($EV$)**: $\$17,540,000$ ($66.19\%$ physical complete)
- **Actual Cost ($AC$)**: $\$23,440,000$
- **Cost Performance Index ($CPI$)**: $0.7483$ ($-25.2\%$ cost efficiency)

#### Detailed $EAC$ Calculations

1. **Standard EVM Forecast ($EAC_1$ - Most Likely)**:

$$EAC_1 = \frac{BAC}{CPI} = \frac{\$26,500,000}{0.74829} = \$35,413,604$$

- **Cost Overrun ($VAC_1$)**: $-\$8,913,604$ ($+33.6\%$ cost overrun).
- **Root Cause**: Unbudgeted subsea structural steel rework ($WBS\text{ }1.3.2$) and crane barge standby rates ($WBS\text{ }1.4.1$).

2. **Floor / Baseline Forecast ($EAC_2$ - Best Case)**:

$$EAC_2 = AC + (BAC - EV) = \$23,440,000 + \$8,960,000 = \$32,400,000$$

- **Cost Overrun ($VAC_2$)**: $-\$5,900,000$ (Freeze current overrun; zero future variance).

3. **Composite Risk Forecast ($EAC_3$ - Worst Case)**:

$$EAC_3 = AC + \frac{BAC - EV}{CPI \times SPI} = \$23,440,000 + \frac{\$8,960,000}{0.7483 \times 0.8556} = \$37,433,522$$

- **Cost Overrun ($VAC_3$)**: $-\$10,933,522$ ($+41.3\%$ cost overrun).

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### 2. Schedule & Completion Date Predictions ($EAC_t$)

#### Time-Based Status Metrics ($ES$ Lipke Theory)

- **Actual Time ($AT$)**: $8.00\text{ Months}$ ($240\text{ Days}$)
- **Earned Schedule ($ES$)**: $7.40\text{ Months}$ ($222\text{ Days}$)
- **Schedule Performance Index ($SPI_t$)**: $0.9250$ ($-7.5\%$ time efficiency)
- **Time Variance ($SV_t$)**: $-0.60\text{ Months} \approx -18.2\text{ Days}$

#### Completion Date Predictions

1. **Earned Schedule Forecast ($EAC_t$ - Time Estimate at Completion)**:

$$EAC_t = \frac{PD}{SPI_t} = \frac{12.00\text{ Months}}{0.9250} = 12.97\text{ Months} \approx 13.0\text{ Months}$$

- **Forecast Finish Date**: **January 31, 2027**
- **Project Delay**: **+31 Calendar Days** ($1.0\text{ Month}$)

2. **Critical Path Schedule Driving Logic**:

- **Driving Task**: $WBS\text{ }1.3.2$ Derrick Tower Mast Assembly (Egersund Yard).
- **Constraint**: Heavy Lift Vessel (*Heerema Sleipnir*) mobilization window is locked from **Sep 15 to Oct 15**.
- **Cascade Risk**: A 30-day delay in mast assembly shifts vessel mobilization into Oct 15–Nov 15, exposing offshore installation to North Sea winter weather downtime ($+$15-20 days severe weather standby risk).

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### 3. To-Complete Performance Index ($TCPI$) Feasibility Analysis

| Baseline Metric | Target | Required Efficiency ($TCPI$) | Feasibility Evaluation | Operational Action |
| --- | --- | --- | --- | --- |
| **Original Budget ($BAC$)** | $\$26.50\text{M}$ | **$2.93$ (or $6.07$ peak)** | **UNVIABLE** | Impossible to achieve. Abandon $BAC$ as financial control baseline. |
| **Revised Forecast ($EAC_1$)** | $\$35.41\text{M}$ | **$1.00$ ($0.75$)** | **ACHIEVABLE** | Standard target. Requires strict cost control on remaining $\$8.96\text{M}$ work. |

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### 4. Key Project Controller Recommendations

1. **Formal Baseline Revision (Re-baseline EAC)**:

- Formally submit a Change Order to the Steering Committee updating the baseline $BAC$ from **$26.50M** to **$35.41M** and extending target completion to **January 31, 2027**.

2. **Yard Rework Clampdown at Egersund ($WBS\text{ }1.3.2$)**:

- Issue a fixed-fee cap on Egersund assembly yard labor. Require 2-shift operations (16 hrs/day) to complete mast assembly by **September 30**.

3. **Heerema Vessel Standby Mitigation ($WBS\text{ }1.4.1$)**:

- Re-negotiate Heavy Lift Mobilization window with Heerema Marine Contractors to avoid $150,000/day standby penalties during weather delays.

4. **Bi-Weekly Earned Schedule ($ES$) Tracking**:

- Track $SPI_t$ bi-weekly. If $SPI_t$ drops below $0.90$, immediately trigger emergency double-shift welding crews.

## 3. Monte Carlo Schedule & Cost Risk Analysis (P50 / P90)

**Project Name**: Offshore EPC Platform Drill Tower Project

**Simulation Iterations**: 10,000 Runs (Triangular Risk Sampling)

**Status Date**: August 31, 2026 (Month 8 / Status Week 36)

**Author**: Frank Ellingsen, Lead Project Controller

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### Executive Summary & P90 Risk Card

A quantitative cost and schedule risk analysis (QCRA / QSRA) was executed using 10,000 Monte Carlo iterations across all 7 Control Accounts and critical path schedule deliverables.

> [!IMPORTANT]

> **P90 Outturn Forecast Summary**:

> - **P90 Outturn Cost**: **$35,815,202** ($-9.32\text{M}$ / $+35.2\%$ variance vs. $26.50\text{M}$ baseline $BAC$).

> - **P90 Contingency Reserve Needed**: **+$401,598** added to current $35.41\text{M}$ deterministic $EAC$.

> - **P90 Completion Date**: **March 14, 2027** ($+72.5\text{ Days}$ / $+2.4\text{ Months}$ schedule delay vs. Dec 31, 2026).

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### 📊 Summary Table of Monte Carlo Risk Percentiles

| Risk Percentile | Confidence Level | Outturn Cost ($EAC$) | Cost Variance vs $BAC$ | Contingency Needed vs $EAC$ | Total Duration | Predicted Completion Date | Schedule Delay vs Dec 31 |
|---|---|---|---|---|---|---|---|
| **P10** | 10% (Optimistic / Low Risk) | **$32,444,302** | -$5,944,302 (-22.4%) | $0.00 | 408.1 Days | **Feb 13, 2027** | +43.1 Days (+1.4 Mos) |
| **P50** | 50% (Median / Expected) | **$34,060,783** | -$7,560,783 (-28.5%) | $0.00 | 422.3 Days | **Feb 27, 2027** | +57.3 Days (+1.9 Mos) |
| **P80** | 80% (Standard Budget Baseline) | **$35,195,026** | -$8,695,026 (-32.8%) | $0.00 | 432.3 Days | **Mar 09, 2027** | +67.3 Days (+2.2 Mos) |
| **P90** | **90% (P90 High Confidence)** | **$35,815,202** | **-$9,315,202 (-35.2%)** | **+$401,598** | **437.5 Days** | **March 14, 2027** | **+72.5 Days (+2.4 Mos)** |
| **P95** | 95% (Extreme Risk Boundary) | **$36,272,986** | -$9,772,986 (-36.9%) | +$859,382 | 441.6 Days | **March 18, 2027** | +76.6 Days (+2.5 Mos) |

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### 1. Simulation Methodology & Input Risk Distributions

The simulation applies triangular probability distributions $(Optimistic, Most\text{ }Likely, Pessimistic)$ to each Control Account:

#### Cost Inputs ($M USD)

- **WBS 1.1 Detail Engineering**: $(3.10, 3.50, 4.20)$
- **WBS 1.2 Procurement**: $(8.20, 9.10, 10.80)$
- **WBS 1.3.1 Verdal Fabrication**: $(4.80, 5.20, 6.00)$
- **WBS 1.3.2 Egersund Mast Rework**: $(5.50, 7.20, 9.50)$ *(Primary Cost Risk)*
- **WBS 1.4.1 Heavy Lift Vessel Mobilization**: $(3.20, 4.50, 6.80)$ *(Primary Weather Standby Risk)*
- **WBS 1.4.2 Topside Lifting & Mating**: $(1.10, 1.40, 2.10)$
- **WBS 1.5 Hook-up & Commissioning**: $(1.70, 1.90, 2.50)$

#### Schedule Duration Inputs (Remaining Days along Critical Path)

- **T106 Derrick Mast Assembly (Egersund)**: $(20, 30, 45\text{ Days})$
- **T107 Heavy Lift Mobilization**: $(25, 30, 50\text{ Days})$
- **T108 Topside Lifting & Mating**: $(25, 31, 45\text{ Days})$
- **T109 Structural Hook-up & NDT**: $(25, 30, 45\text{ Days})$
- **T110 Pre-Commissioning & Handover**: $(35, 47, 65\text{ Days})$

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### 2. Risk Driver Sensitivity & Tornado Analysis

1. **Egersund Assembly Yard Rework ($WBS\text{ }1.3.2$)**: Accounts for **48.2% of total cost variance**. Structural pipe tolerance issues require 24/7 NDT inspection and specialized welding crews.

2. **Offshore Weather Standby ($WBS\text{ }1.4.1$)**: Accounts for **36.5% of total schedule variance**. Shifting vessel mobilization from September into October/November exposes the heavy lift crane barge (*Heerema Sleipnir*) to North Sea sea-state limits ($Hs > 2.5m$).

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### 3. Project Controller Recommendations for Steering Committee

1. **Establish P80 / P90 Financial Reserve**:

- Approve a revised Control Budget of **$35.82M** ($P90$) by allocating a **+$401,598 management contingency reserve** on top of the $35.41\text{M}$ deterministic $EAC$.

2. **Schedule Target & Buffer**:

- Establish **March 14, 2027** as the P90 contractual handover date for offshore commissioning.

3. **CSV Export**:

- Simulation raw dataset saved to [`03_power_bi/Monte_Carlo_Simulation_Results.csv`](file:///C:/Users/frank/Desktop/EVM/03_power_bi/Monte_Carlo_Simulation_Results.csv).

## 4. EVM Variance Explanations & Corrective Action Audit

**Project:** North Sea Oil Platform Drill Tower Construction ($26.5M EPC Program)

**Status Date:** August 31, 2026 (Month 8 / Status Date)

**Target Role:** Project Controller (Frank Ellingsen) / Executive Steering Committee

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### 📌 Executive Summary & Red Indicator Audit

At Status Month 8, the North Sea Drill Tower program exhibits critical cost overruns ($CPI = 0.75$) and critical path schedule delays ($+31\text{ Days}$, $SPI = 0.86$). The table below outlines the root cause explanations and recommended corrective action plans for all active red performance indicators.

| Red Indicator | Status Metric | Threshold | Root Cause Explanation | Recommended Actionable Next Steps | Accountable CAM |
|---|---|---|---|---|---|
| **Cost Variance ($CV$)** | **-$5,900,000** | $CPI = 0.75$ ($< 0.90$) | Structural steel fitting rework, unbudgeted welding labor hours, and premium overtime rates at Verdal and Egersund yards. | Enforce Design Change Freeze on WBS 1.3 deliverables. Convert open T&M subcontracts to fixed unit-rate capped milestones. | O. Eriksen |
| **Outturn Deficit ($VAC$)** | **-$8,913,604** | $EAC = \$35.41\text{M}$ | Linear extrapolation of poor historical cost performance ($CPI = 0.75$) indicates $33.6\%$ budget overrun against $BAC$. | Submit formal EAC re-baseline proposal to Executive Steering Committee. Request $3.5\text{M}$ drawdown from Management Reserve. | Frank Ellingsen |
| **Target Efficiency ($TCPI_{BAC}$)** | **6.07 (BAC)** | Unviable ($> 1.10$) | Original $BAC$ budget target requires impossible cost efficiency of $607\%$ for remaining work. | Formally transition operational target to $TCPI_{EAC} = 1.00$. Enforce weekly EVM gate reviews for CAMs with $TCPI_{EAC} > 1.05$. | Frank Ellingsen |
| **Schedule Slippage ($SV$)** | **-$2,960,000** | $+31\text{ Days}$ Delay | Procurement delay on Tubular Steel (T103) cascaded to Verdal (T105) and Egersund (T106), threatening Heavy Lift window. | Fast-track Egersund rigging with parallel work-fronts. Re-sequence pre-commissioning tasks. Negotiate 10-day vessel window extension. | M. Berg / K. Solberg |
| **WBS 1.3.2 Mast Assembly** | **-$2,330,000** | $CPI = 0.57$ | Egersund rigging yard productivity bottleneck; high sub-tier contractor churn and excessive rework hours. | Deploy resident Project Controller to Egersund full-time. Audit daily labor timecards and enforce weld quality gates. | O. Eriksen |

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### 🔍 Detailed Variance Analysis & Recommended Actions

#### 1. 🔴 Cost Variance ($CV = -\$5,900,000$, $CPI = 0.75$)

- **Root Cause Explanation**: Actual expenditure ($AC = \$23.44\text{M}$) severely exceeds Earned Value ($EV = \$17.54\text{M}$). Major cost growth originated in WBS 1.3.1 (Verdal Yard Sub-Structure Fabrication, $CV = -\$850\text{k}$) and WBS 1.3.2 (Egersund Derrick Mast Assembly, $CV = -\$2.33\text{M}$). Key drivers include engineering change notices during fabrication, out-of-sequence welding, and high overtime premiums.
- **Recommended Action Plan**:

1. **Design Change Freeze**: Formally freeze all pending Engineering Change Requests (CRs) for WBS 1.3 until offshore installation completion.

2. **Subcontract Conversion**: Convert open Time & Materials (T&M) subcontracts with yard fabrication suppliers into fixed unit-rate capped milestones.

3. **On-site Controller Audit**: Station a dedicated Project Controller at Egersund yard to approve daily man-hour timecards and verify physical milestone completion before payment release.

### 2. 🔴 Outturn Forecast Deficit ($VAC = -\$8,913,604$, $EAC = \$35,413,604$)

- **Root Cause Explanation**: Extrapolating historical efficiency ($CPI = 0.75$) yields a cumulative $EAC_{CPI}$ of $\$35.41\text{M}$ against the baseline $BAC$ of $\$26.50\text{M}$, creating an outturn deficit of $\$8.91\text{M}$.
- **Recommended Action Plan**:

1. **EAC Re-baseline Proposal**: Prepare an executive re-baseline submission for the Steering Committee to adjust the project budget baseline to $\$35.41\text{M}$.

2. **Management Reserve (MR) Drawdown**: Formally request a $\$3.50\text{M}$ allocation from corporate Management Reserve to absorb non-recoverable raw material price inflation.

3. **ETC Ceiling Enforcement**: Set strict Estimate-to-Complete ($ETC = \$11.97\text{M}$) expenditure caps per WBS control account.

### 3. 🔴 Target Cost Efficiency ($TCPI_{BAC} = 6.07$)

- **Root Cause Explanation**: $TCPI_{BAC} = \frac{BAC - EV}{BAC - AC} = \frac{26.50 - 17.54}{26.50 - 23.44} = 6.07$. Achieving the remaining work within the original $\$26.50\text{M}$ budget is mathematically unviable, requiring $607\%$ cost efficiency.
- **Recommended Action Plan**:

1. **Pivot Operational Metric**: Cease using $TCPI_{BAC}$ as an operational performance target.

2. **Enforce $TCPI_{EAC}$ Tracking**: Monitor remaining work against $TCPI_{EAC} = \frac{BAC - EV}{EAC - AC} = \frac{26.50 - 17.54}{35.41 - 23.44} = 1.00$.

3. **Weekly CAM Reviews**: Implement mandatory weekly EVM variance gate reviews for any Control Account Manager whose $TCPI_{EAC}$ exceeds $1.05$.

### 4. 🟡/🔴 Critical Path Schedule Delay ($SV = -\$2,960,000$, $+31\text{ Days}$)

- **Root Cause Explanation**: Procurement delays on High-Grade Tubular Steel (T103, $+31$ days) delayed Verdal Sub-Structure Fabrication (T105) and Egersund Derrick Assembly (T106). This pushes Heavy Lift Mobilization (T107) into late September/October, risking autumn North Sea weather windows.
- **Recommended Action Plan**:

1. **Rigging Fast-Tracking**: Parallelize structural mast assembly work-fronts at Egersund by deploying additional heavy lift cranes.

2. **Pre-Commissioning Re-sequencing**: Shift 14 days of pre-commissioning testing (T110) post-mating to shorten the critical path prior to vessel mobilization.

3. **Marine Fleet Contingency**: Negotiate a 10-day weather-window extension agreement with Heerema Marine Contractors to prevent standby penalty charges ($150\text{k}/day$).

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### 🌐 Web Dashboard Integration

The interactive web dashboard at [`04_dashboard/index.html`](file:///C:/Users/frank/Desktop/EVM/04_dashboard/index.html) and [`04_dashboard/drill_tower_web_report.html`](file:///C:/Users/frank/Desktop/EVM/04_dashboard/drill_tower_web_report.html) features:

- **Red Indicator Audit Table**: Direct table highlighting root causes, actions, and accountable CAMs.
- **Interactive Badges & Audit Modal**: Clicking any red status metric badge launches an interactive modal popup with detailed root causes, step-by-step action plans, and accountable owner contact info.
- **S-Curve Dynamic Layer Toggles**: Allows switching between Cost Variance ($CV$), Schedule Variance ($SV$), and Earned Schedule ($ES$).

## 5. Monthly Cash Burn Speed & Budget Depletion (Burn Out)

**Project Name**: Offshore EPC Platform Drill Tower Project

**Status Date**: August 31, 2026 (Month 8 / Status Week 36)

**Author**: Frank Ellingsen, Lead Project Controller

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### 🎯 Executive Burn Rate & Cash Runway Summary

| Burn Rate Metric | Historical Value (Months 1-8) | Forecast Value (Months 9-13) | Benchmark Target | Evaluation & Impact |
|---|---|---|---|---|
| **Average Monthly Burn Rate** | **$2,930,000/month** | **$2,394,721/month** | $2,208,333/month | **EXPEDITED BURN**. Spending 32.7% faster than baseline monthly plan. |
| **Peak Monthly Spend** | **$4,450,000/month** (Jun) | **$3,300,000/month** (Oct) | $4,000,000/month | High yard labor density & vessel standby daily rates. |
| **Cumulative Actual Spend** | **$23,440,000** | **$35,413,604** (EAC) | $20,500,000 (PV) | **-$5.90M Cost Overrun to Date**. |
| **Budget Burn Out Month** | **Month 9 (September 2026)** | - | Month 12 (Dec 2026) | **RUNWAY EXHAUSTED**. 100% of baseline budget ($BAC$) depleted in Month 9. |
| **Outturn Overrun Cash Gap** | **+$8,913,604** | - | $0.00 | Capital overrun required to reach completion in Month 13. |

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### 📊 Monthly Burn Rate & Cumulative Expenditure Table

| Month | Date | PV Monthly | EV Monthly | AC Monthly | Cumulative PV | Cumulative EV | Cumulative AC | Budget Status / Burn Out Runway |
|---|---|---|---|---|---|---|---|---|
| **M1** | Jan 31 | $300,000 | $300,000 | $320,000 | $300,000 | $300,000 | $320,000 | Baseline Runway ($26.18M remaining) |
| **M2** | Feb 28 | $900,000 | $840,000 | $950,000 | $1,200,000 | $1,140,000 | $1,270,000 | Baseline Runway ($25.23M remaining) |
| **M3** | Mar 31 | $2,700,000 | $2,260,000 | $2,900,000 | $3,900,000 | $3,400,000 | $4,170,000 | Baseline Runway ($22.33M remaining) |
| **M4** | Apr 30 | $3,800,000 | $3,520,000 | $4,050,000 | $7,700,000 | $6,920,000 | $8,220,000 | Baseline Runway ($18.28M remaining) |
| **M5** | May 31 | $3,000,000 | $2,900,000 | $3,220,000 | $10,700,000 | $9,820,000 | $11,440,000 | Baseline Runway ($15.06M remaining) |
| **M6** | Jun 30 | $4,000,000 | $2,750,000 | $4,450,000 | $14,700,000 | $12,570,000 | $15,890,000 | Baseline Runway ($10.61M remaining) |
| **M7** | Jul 31 | $3,000,000 | $2,620,000 | $3,850,000 | $17,700,000 | $15,190,000 | $19,740,000 | Baseline Runway ($6.76M remaining) |
| **M8** | Aug 31 | $2,800,000 | $2,350,000 | $3,700,000 | $20,500,000 | $17,540,000 | **$23,440,000** | **CRITICAL RUNWAY** ($3.06M remaining) |
| **M9** | Sep 30 | $2,500,000 | $2,100,000 | **$3,200,000** | $23,000,000 | $19,640,000 | **$26,640,000** | 💥 **BUDGET BURN OUT** ($BAC $26.5M Depleted!) |
| **M10** | Oct 31 | $1,500,000 | $2,200,000 | **$3,300,000** | $24,500,000 | $21,840,000 | **$29,940,000** | Overrun Phase (+$3.44M Over budget) |

| Month | Date | PV Monthly | EV Monthly | AC Monthly | Cumulative PV | Cumulative EV | Cumulative AC | Budget Status / Burn Out Runway |
|---|---|---|---|---|---|---|---|---|
| **M11** | Nov 30 | $1,000,000 | $2,000,000 | **$2,500,000** | $25,500,000 | $23,840,000 | **$32,440,000** | Overrun Phase (+$5.94M Over budget) |
| **M12** | Dec 31 | $1,000,000 | $1,800,000 | **$1,800,000** | $26,500,000 | $25,640,000 | **$34,240,000** | Overrun Phase (+$7.74M Over budget) |
| **M13** | Jan 31 | $0 | $860,000 | **$1,173,604** | $26,500,000 | $26,500,000 | **$35,413,604** | **FINAL COD** ($EAC Outturn Reached) |

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### 📈 Visual Burn Rate Dynamics

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CUMULATIVE CASH BURN ($M)

$36M +--------------------------------------------------------------------------* (Final EAC: $35.41M)

$32M | *--------' (Overrun Expenditure)

$28M | *--------'

$26.5M+----------------------------------------------- 💥 BURN OUT------------------ (BAC Budget Ceiling)

$24M | *--------'

$20M | *--------'

$16M | *--------'

$12M | *--------'

$8M | *--------'

$4M | *--------'

$0 +--+--------+--------+--------+--------+--------+--------+--------+--------+--------+

M1 M3 M5 M7 M8(Now) M9(Burn) M11 M13(COD)

```

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### 💡 Key Project Controller Recommendations

1. **Secure Overrun Credit Line Before Month 9**:

- The project will exhaust its $26.50M baseline budget in **September 2026 (Month 9)**. Emergency liquidity approval of **$8.91M** must be finalized prior to September 15.

2. **Cap Monthly Burn Speed at Egersund Yard**:

- Monthly actual spend averaged **$3.78M/month** in Months 6-8. Impose a monthly spending limit of **$3.00M/month** for Q4 yard operations.

## 6. Cost Variance Waterfall Bridge ($BAC to EAC)

**Project Name**: Offshore EPC Platform Drill Tower Project

**Status Date**: August 31, 2026 (Month 8 / Status Week 36)

**Author**: Frank Ellingsen, Lead Project Controller

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### 🎯 Executive Cost Variance Waterfall Bridge Summary

The **Cost Variance Waterfall Bridge** reconciles the step-by-step progression from the original **Baseline Budget ($BAC = \$26,500,000$)** to the final **Outturn Forecast ($EAC_1 = \$35,413,604$)**, isolating the exact financial impact of cost variances ($VAC$) across each WBS element and time-based overhead extension.

> [!IMPORTANT]

> **Total Project Variance**: **+$8,913,604** (+33.6% cost expansion).

> **Primary Cost Driver**: Egersund Derrick Mast Assembly Rework (WBS 1.3.2) contributing **$2.40M (26.9%)** of total cost expansion.

> **Secondary Cost Driver**: Earned Schedule Time Delay & Extended Overhead (SPI_t = 0.9250) contributing **$3.81M (42.8%)** of outturn escalation.

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### 📊 Step-by-Step Waterfall Cost Bridge Table

| Bridge Step | WBS Code | Control Account / Variance Component | Incremental Cost Impact | Cumulative Project Cost | % Contribution to Total Overrun | Primary Root Cause & Description |
|---|---|---|---|---|---|---|
| **0** | `1.0` | **Original Baseline Budget ($BAC$)** | **$26,500,000** | **$26,500,000** | - | Original approved EPC contract target budget. |
| **1** | `1.1.1` | Detail Structural Engineering | **+$300,000** | $26,800,000 | 3.4% | Redesign of connection plates & third-party DNV class review. |
| **2** | `1.2.1` | Procurement & Tubular Steel | **+$600,000** | $27,400,000 | 6.7% | Raw subsea tubular steel price inflation & mud pump shipping surcharge. |
| **3** | `1.3.1` | Verdal Sub-Structure Fabrication | **+$200,000** | $27,600,000 | 2.2% | Minor yard overtime for weld inspection. |
| **4** | `1.3.2` | **Egersund Derrick Mast Rework** | **+$2,400,000** | **$30,000,000** | **26.9%** | **PRIMARY REWORK DRIVER**: Pipe tolerance mismatch requiring 24/7 NDT welding. |
| **5** | `1.4.1` | Heavy Lift Vessel Standby | **+$1,300,000** | $31,300,000 | 14.6% | Marine weather standby daily rates (*Heerema Sleipnir*). |
| **6** | `1.4.2` | Offshore Topside Lifting & Mating | **+$100,000** | $31,400,000 | 1.1% | Pre-lift trial fitting and alignment fixtures. |
| **7** | `1.5.1` | Hook-up & Commissioning | **+$200,000** | $31,600,000 | 2.2% | Extended offshore commissioning specialist crew. |
| **8** | `EAC_t` | **Schedule Delay Overhead Extension** | **+$3,813,604** | **$35,413,604** | **42.8%** | **TIME DELAY SPREAD**: +31-day schedule delay ($SPI_t = 0.9250$) spreading PMO & site overhead. |
| **END** | `1.0` | **Final Forecast Outturn ($EAC$)** | **+$8,913,604** | **$35,413,604** | **100.0%** | Final predicted cost outcome at completion. |

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📈 Visual Waterfall Cost Progression

```

BASE $26.50M (BAC)

│

├───[+$0.30M] Step 1: Detail Engineering Revisions

│

├───[+$0.60M] Step 2: Tubular Steel Procurement Inflation

│

├───[+$0.20M] Step 3: Verdal Sub-Structure Fabrication

│

├───[+$2.40M] 💥 Step 4: Egersund Derrick Mast Yard Rework (MAJOR YARD DRIVER)

│

├───[+$1.30M] ⚓ Step 5: Heavy Lift Vessel Weather Standby

│

├───[+$0.10M] Step 6: Topside Mating Fit-up

│

├───[+$0.20M] Step 7: Hook-up Commissioning Crew

│

└───[+$3.81M] ⏳ Step 8: Time Delay Overhead Spread (SCHEDULE IMPACT)

│

OUTTURN $35.41M (Final EAC)

```

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💡 Steering Committee Control Takeaways

1. **Yard Rework Containment**:

- $WBS\text{ }1.3.2$ represents $26.9\%$ of total overrun. Enforce strict fixed-fee caps on remaining yard welding hours.

2. **Schedule Recovery to Limit Overhead Spread**:

- $42.8\%$ of total overrun stems from time-based project management & site overhead extension ($+\$3.81\text{M}$). Recovering 15 calendar days reduces overhead by $\approx \$1.80\text{M}$.

## 7. EPC Executive Risk Matrix & Mitigation Register

**Project Name**: Offshore EPC Platform Drill Tower Project

**Status Date**: August 31, 2026 (Month 8 / Status Week 36)

**Author**: Frank Ellingsen, Lead Project Controller

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### 🎯 Executive Risk Heatmap Matrix (5x5 Grid)

The 5x5 Risk Heatmap Matrix plots project risks by **Probability (Likelihood 1-5)** versus **Impact (Severity 1-5)**. Total Risk Score = $Probability \times Impact$.

```

PROBABILITY

(5) Almost Certain | [5] | [10] | [15] | [20] | [25] |

(4) High | [4] | [8] | [12] | 🚨 R02 | 🚨 R01 | <-- CRITICAL ZONE

(3) Moderate | [3] | [6] | ⚠️ R04 | ⚠️ R03 | [15] |

(2) Low | [2] | [4] | 🟢 R05 | [8] | [10] |

(1) Rare | [1] | [2] | [3] | [4] | [5] |

+--------+--------+--------+--------+--------+

(1) (2) (3) (4) (5)

Negligible Minor Moderate Major Critical

IMPACT

```

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### 📋 Comprehensive Risk Register & Mitigation Action Plan

| Risk ID | Risk Category | Risk Event Description | WBS | Risk Owner (CAM) | Prob (1-5) | Imp (1-5) | Risk Score | Risk Exposure ($) | Actionable Mitigation Strategy | Status |
|---|---|---|---|---|---|---|---|---|---|---|
| **R01** | **Fabrication** | **Egersund Mast Assembly Dimensional Out-of-Tolerance & Yard Labor Rework** | 1.3.2 | O. Eriksen | 4 | 5 | 20 (Critical) | **$2,400,000** | Deploy 24/7 dual-shift NDT welding specialists; cap yard labor billing under a fixed-fee agreement. | **Active** |
| **R02** | **Offshore Marine** | **Heavy Lift Vessel Mobilization Delay & North Sea Weather Standby** | 1.4.1 | K. Solberg | 4 | 4 | 16 (Critical) | **$1,800,000** | Negotiate flexible weather window with Heerema; monitor 7-day wave height ($Hs < 2.5m$). | **Active** |
| **R03** | **Procurement** | **High-Grade Subsea Tubular Steel | 1.2.1 | M. Berg | 3 | 4 | 12 (High) | **$1,200,000** | Dual-source tubular steel from | **Monitored** |

| Risk ID | Risk Category | Risk Event Description | WBS | Risk Owner (CAM) | Prob (1-5) | Imp (1-5) | Risk Score | Risk Exposure ($) | Actionable Mitigation Strategy | Status |
|---|---|---|---|---|---|---|---|---|---|---|
| | | Mill Delivery Lags & Price Inflation** | | | | | | | European backup mills; arrange expedited hot-shot freight. | |
| **R04** | **Offshore Hook-Up** | **Topside Lifting & Mating Mechanical Interface Interference** | 1.4.2 | T. Nygård | 3 | 3 | 9 (Medium) | **$600,000** | Perform 3D laser scan trial fit at Egersund yard prior to offshore mobilization. | **Monitored** |
| **R05** | **Engineering** | **Structural Detail Engineering Interface Errors & AFC Revisions** | 1.1.1 | H. Lindqvist | 2 | 3 | 6 (Low) | **$300,000** | Enforce 100% 3D CAD clash detection and third-party DNV class design verification. | **Closed** |

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### 💰 Quantitative Financial Risk Summary

- **Total Unmitigated Financial Exposure**: **$6,300,000**
- **Post-Mitigation Expected Monetary Value (EMV)**: **$3,410,000**
- **Recommended Management Risk Reserve**: **$3,500,000** (Aligned with P90 Monte Carlo simulation reserve of $+\$401,598$ above $35.41\text{M}$ outturn).

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### 🛡️ Risk Governance & Escalation Triggers

1. **Red Risk Escalation (Score $\ge 15$)**: Weekly review with Project Director & CFO. Immediate CAM intervention required.

2. **Amber Risk Monitoring (Score $8 - 14$)**: Bi-weekly review at Project Control Meetings.

3. **Green Risk Tracking (Score $\le 7$)**: Monthly review by CAMs.

## 8. Vertical Swimlane WBS Control Account Breakdown

**Project Name**: Offshore EPC Platform Drill Tower Project

**Status Date**: August 31, 2026 (Month 8 / Status Week 36)

**Author**: Frank Ellingsen, Lead Project Controller

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### 🎯 Executive Category Breakdown Summary

| Category / Discipline | WBS Code | Control Account Manager (CAM) | Baseline Budget ($BAC$) | Outturn Forecast ($EAC$) | Cost Variance ($VAC$) | Physical Progress (%) | Current Status |
|---|---|---|---|---|---|---|---|
| **1. Engineering & Design** | `1.1` | H. Lindqvist | **$3,200,000** | **$3,500,000** | -$300,000 | **100%** | Achieved |
| **2. Procurement & Supply Chain** | `1.2` | M. Berg | **$8,500,000** | **$9,100,000** | -$600,000 | **94.1%** | Advanced |
| **3. Yard Fabrication & Assembly** | `1.3` | O. Eriksen | **$9,800,000** | **$12,400,000** | -$2,600,000 | **77.6%** | Critical Yard Rework |
| **4. Offshore Installation & Marine** | `1.4` | K. Solberg | **$5,000,000** | **$6,400,000** | -$1,400,000 | **19.2%** | Standby Risk |
| **5. Hook-up & Commissioning** | `1.5` | T. Nygård | **$1,800,000** | **$2,000,000** | -$200,000 | **0.0%** | Scheduled Q4 |
| **TOTAL PROJECT** | `1.0` | **Frank Ellingsen** | **$26,500,000** | **$35,413,604** | **-$8,913,604** | **66.2%** | **Overrun & Delayed** |

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### 🏛️ Vertical Swimlane Work Breakdown Structure (WBS)

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+--------------------+ +--------------------+ +--------------------+ +--------------------+ +--------------------+

| 1.1 ENGINEERING | 1.2 PROCUREMENT | 1.3 FABRICATION | 1.4 INSTALLATION | 1.5 COMMISSIONING |
|---|---|---|---|---|

+--------------------+ +--------------------+ +--------------------+ +--------------------+ +--------------------+

| 1.1.1 Detail Eng | 1.2.1 Steel Pipes | 1.3.1 Verdal Sub | 1.4.1 Heavy Lift | 1.5.1 Hook-up & NDT | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| BAC: $1.80M | 100% | BAC: $5.00M | 100% | BAC: $5.00M | 90% | BAC: $3.20M | 30% | BAC: $1.00M | 0% |
| EAC: $2.10M (-$300K) | EAC: $5.50M (-$500K) | EAC: $5.20M (-$200K) | EAC: $4.50M (-$1.3M) | EAC: $1.10M (-$100K) | | | | | |

+--------------------+ +--------------------+ +--------------------+ +--------------------+ +--------------------+

| 1.1.2 Piping Design | 1.2.2 Mud Pumps | 1.3.2 Egersund Mast | 1.4.2 Topside Mating | 1.5.2 Handover COD | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| BAC: $1.40M | 100% | BAC: $3.50M | 85% | BAC: $4.80M | 65% | BAC: $1.80M | 0% | BAC: $0.80M | 0% |
```

| 1.1.2 Piping Design | 1.2.2 Mud Pumps | 1.3.2 Egersund Mast | 1.4.2 Topside Mating | 1.5.2 Handover COD |
| --- | --- | --- | --- | --- |
| EAC: $1.40M ($0K) | EAC: $3.60M (-$100K) | EAC: $7.20M (-$2.4M) | EAC: $1.90M (-$100K) | EAC: $0.90M (-$100K) |

+---------------------+ +---------------------+ +---------------------+ +---------------------+ +---------------------+

```

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### 🔍 Discipline Performance Insights

1. **Engineering Swimlane ($1.1$)**: Completed 100% physical progress with a minor $-\$300\text{K}$ cost variance due to design revisions on structural steel connection plates.

2. **Procurement Swimlane ($1.2$)**: $94.1\%$ complete. Steel tubular procurement encountered $-\$500\text{K}$ raw material price inflation.

3. **Fabrication Swimlane ($1.3$)**: Primary project cost overrun center ($-\$2.60\text{M}$ variance). Verdal yard sub-structure is $90\%$ complete, while Egersund mast assembly requires extended NDT rework ($-\$2.40\text{M}$ overrun).

4. **Installation Swimlane ($1.4$)**: Heavy lift vessel mobilization has incurred $-\$1.30\text{M}$ standby cost escalation due to North Sea autumn weather windows.

5. **Commissioning Swimlane ($1.5$)**: Scheduled for Q4 2026 / Q1 2027 following completion of topside mating.
```

## 9. Project Financial Metrics & Appraisal

**Project Name**: Offshore EPC Platform Drill Tower Project

**CAPEX Baseline ($BAC$)**: $26,500,000

**CAPEX Outturn Forecast ($EAC$)**: $35,413,604

**Discount Rate / Hurdle Rate ($WACC$)**: 10.0%

**Author**: Frank Ellingsen, Financial Controller & BI Specialist

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### Executive Summary & Financial Ratio Card

| Capital Appraisal Metric | Standard Formula | Outturn Value | Threshold Target | Commercial Decision & Evaluation |
|---|---|---|---|---|
| **Net Present Value ($NPV$)** | $\sum \frac{CF_t}{(1+r)^t} - CAPEX$ | **+$14,899,563** | $> \$0.00$ | **ACCEPT**. Project creates $14.90M net shareholder value above 10% WACC. |
| **Internal Rate of Return ($IRR$)** | $NPV(IRR) = 0$ | **18.86%** | $> 10.0\%$ WACC | **EXCEEDS HURDLE**. Premium spread of +886 bps over corporate cost of capital. |
| **Profitability Index ($PI$)** | $\frac{PV(\text{Inflows})}{CAPEX}$ | **1.42** | $> 1.00$ | **STRONG BENCHMARK**. Generates $1.42 in present value per $1.00 spent. |
| **Simple Payback Period** | $\frac{CAPEX}{\text{Annual } CF}$ | **4.43 Years** | $< 5.00$ Years | **VIABLE**. Capital recovered in 53.1 months (May 2031). |
| **Total Cumulative ROI** | $\frac{\text{Net Cash Flow}}{CAPEX}$ | **134.37%** | $> 50.0\%$ | **EXCELLENT**. Total net cash inflows of $75.00M on $35.41M investment. |
| **Annualized ROI (CAGR)** | $(1 + ROI)^{1/N} - 1$ | **8.89%/year** | $> 7.0\%$/year | **STABLE**. Sustained annual compounded return across 10-year asset life. |
| **Future Value (FV at Yr 10)** | $\sum CF_t \cdot (1+r)^{10-t}$ | **$130,499,397** | - | Gross compounded terminal cash value generated by Year 10 (2036). |
| **Net Future Value ($NFV$)** | $FV(\text{Inflows}) - FV(CAPEX)$ | **+$38,645,628** | $> \$0.00$ | Net value created in Year 10 dollar terms. |

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### 1. Cash Flow Schedule & Capital Outlay

#### Capital Expenditure ($CAPEX$)

- **Original Baseline ($BAC$)**: $\$26,500,000$
- **Outturn Forecast ($EAC_1$)**: $\$35,413,604$ (Includes $\$8.91\text{M}$ cost overrun from Egersund yard rework and Heerema vessel standby).

#### Operational Inflows ($Years\text{ }1 - 10$)

- **Daily Rig Revenue Addition**: $\$110,000/\text{day} \times 365 \times 85\% \text{ utilization} = \$34,127,500/\text{year}$
- **Annual Operational Expenditure ($OPEX$)**: $\$26,127,500/\text{year}$
- **Net Annual Cash Flow ($CF_t$)**: **$\$8,000,000/\text{year}$**
- **Terminal Salvage Value (Year 10)**: $\$3,000,000$

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### 2. Sensitivity Analysis (Cost Overrun Impact on $NPV$ & $IRR$)

| CAPEX Scenario | Total Outturn ($EAC$) | Net Present Value ($NPV$) | Internal Rate of Return ($IRR$) | Payback Period | Profitability Index ($PI$) |
|---|---|---|---|---|---|
| **Baseline Plan ($BAC$)** | $\$26.50\text{M}$ | **+$23,813,167** | **27.42%** | **3.31 Years** | **1.90** |
| **Best Case ($EAC_2$)** | $\$32.40\text{M}$ | **+$17,913,167** | **21.28%** | **4.05 Years** | **1.55** |
| **Most Likely ($EAC_1$)** | **$\$35.41\text{M}$** | **+$14,899,563** | **18.86%** | **4.43 Years** | **1.42** |
| **Worst Case ($EAC_3$)** | $\$37.43\text{M}$ | **+$12,879,645** | **17.38%** | **4.68 Years** | **1.34** |

> [!NOTE]

> Even under the Worst Case Cost Overrun scenario ($EAC_3 = \$37.43\text{M}$), the project remains robustly profitable with an **$NPV$ of +$12.88M** and an **$IRR$ of 17.38%**, well clear of the **10.0% corporate hurdle rate**.